

Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

Technical Requirement ID: TR-E2E-001

Related Functional Requirement(s): FR-E2E-001, FR-E2E-002, FR-E2E-003, FR-E2E-004, FR-E2E-005, FR-E2E-006

Objective: Implement a data pipeline to extract data from TXDB, transform and consolidate it, and load it into the STG_E2E_BC_K2H_TXDB_DATA Oracle staging table.

Target Cluster Configuration:

Cluster Name: Oracle Data Integration Cluster

Databricks Runtime Version: 7.3.x-scala2.12

Node Type: Standard_DS3_v2

Driver Node: Standard_DS3_v2

Worker Nodes: 2-4 Standard_DS3_v2

Autoscaling: Enabled

Auto Termination: 30 minutes

Libraries Installed: Oracle JDBC driver, Spark Oracle connector

Source Data Details:

Dataset Name: T_TX_REQUEST_POLICY, T_TX_BASIC, T_TX_RELATION, T_TX_DTL_BENEFICIARY_CHANGE

Location: TXDB database

Format: Oracle database tables

Description: Source data for beneficiary change information

Target Data Details:

Output Name: STG_E2E_BC_K2H_TXDB_DATA

Location: Oracle staging table

Format: Oracle database table

Description: Consolidated and transformed beneficiary change data

Job Flow / Pipeline Stages:

- Extract data from TXDB source tables
- Transform and consolidate data
- Load data into STG_E2E_BC_K2H_TXDB_DATA Oracle staging table

Data Transformations / Business Logic:**Step: Standardize and trim name fields**

Description: Apply LTRIM/RTRIM to Latin and Kanji name fields and default Kanji fields to empty string if null/blank

Transformation Logic: Use Spark SQL to apply LTRIM/RTRIM and handle null/blank values

Step: Normalize Kanji characters

Description: Validate Kanji characters against Windows-932 (Shift-JIS) code ranges and replace out-of-range characters with U+25A0

Transformation Logic: Use Spark SQL to validate Kanji characters and replace out-of-range characters

Step: Handle date fields

Description: Cast date fields to Oracle DATE

Transformation Logic: Use Spark SQL to cast date fields to Oracle DATE

Error Handling and Logging:

- Log errors and exceptions to a log file
- Implement retry mechanism for database connections and data loading
- Use Spark's built-in error handling mechanisms to handle data transformation errors